

ECE250: Signals & Systems Quiz-5

Date: 8th October, 2025

Total Marks: 10

Duration: 15 Minutes

Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
 2. Institute Plagiarism policy are strictly applicable.
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[CO1 CO2 CO3] [Q1]: Consider a causal continuous-time LTI system whose input $x(t)$ and output $y(t)$ are related by the following differential equation:

$$2\frac{d}{dt}y(t) + 3y(t) = x(t). \quad (1)$$

Find the Fourier series representation of the output $y(t)$ for each of the following inputs:

(a) $x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - n)$ **[5 Marks]**

(b) $x(t) = \sum_{n=-\infty}^{+\infty} (-1)^n \delta(t - n)$ **[5 Marks]**